

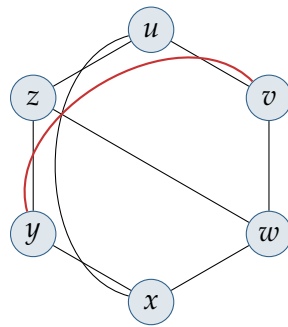
Planarity

Theorem 1

The graphs K_5 and $K_{3,3}$ are non-planar.

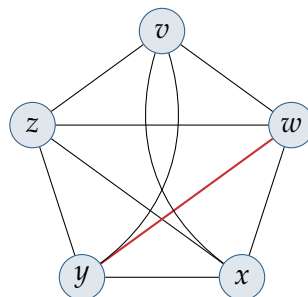
Proof. Suppose first that $K_{3,3}$ is planar. Since $K_{3,3}$ has a cycle $u \rightarrow v \rightarrow w \rightarrow x \rightarrow y \rightarrow z \rightarrow u$ of length 6, any planar drawing must contain this cycle drawn in the form of a hexagon.

Now the edge wz must lie either wholly inside the hexagon or wholly outside it. We deal with the case in which wz lies inside the hexagon — the other case is similar. Since the edge ux must not cross the edge wz , it must lie outside the hexagon. It is then impossible to draw the edge vy , as it would cross either ux or wz .



Now suppose K_5 is planar. Since K_5 has a cycle $v \rightarrow w \rightarrow x \rightarrow y \rightarrow z \rightarrow v$ of length 5, any planar drawing must contain this cycle drawn in the form of a pentagon.

Now the edge wz must lie either wholly inside the pentagon or wholly outside it. We deal with the case in which wz lies inside the pentagon — the other case is similar. Since the edges vx and vy must not cross the edge wz , they must both lie outside the pentagon. But the edge xz cannot cross vy , and so must lie inside the pentagon; similarly, wy cannot cross vx , and so must also lie inside. Then wy and xz must cross, a contradiction.



□

Theorem 2 (Kuratowski, 1930)

A graph is planar if and only if no subgraph is homeomorphic to K_5 or $K_{3,3}$.

Theorem 3

A graph is planar if and only if no subgraph is contractible to K_5 or $K_{3,3}$.

Handshaking Lemma (for Faces)

In any plane drawing of a planar graph,

$$\sum \deg(f_i) = 2 |E(G)|.$$

Proof. In a plane drawing, each edge has two sides (two incident faces):

- the edge may lie on the boundary of a single face (counted twice), or
- the edge lies on the boundary of a face f_1 on one side, and a face f_2 on the other side.

In either case, each edge contributes exactly 2 to the total degree. Hence $\sum \deg(f_i) = 2 |E(G)|$. $\square$

Theorem 4 (Euler, 1750)

Let G be a plane drawing of a connected planar graph, and let n , m , and F denote respectively the number of vertices, edges, and faces of G . Then

$$n - m + F = 2.$$

Proof. By induction on the number of edges of G . If $m = 0$, then $n = 1$, $F = 1$. The theorem is true in this case.

Now, suppose that the theorem holds for all graphs with at most $m - 1$ edges. Let G be a graph with m edges. If G is a tree, then $m = n - 1$, $F = 1$, so that indeed $n - m + F = 2$ as required.

If G is not a tree, let e be an edge in some cycle of G . Then $G - e$ is a connected plane graph with n vertices, $m - 1$ edges, $F - 1$ faces, so that $n - (m - 1) + (F - 1) = 2$ by the induction hypothesis. Adding back the removed edge gives $n - m + F = 2$, as required. $\square$

Corollary 1

- (i) If G is a connected simple planar graph with n (≥ 3) vertices and m edges, then $m \leq 3n - 6$.
- (ii) If, in addition, G has no triangle, then $m \leq 2n - 4$.

Proof. (i): Since each face is bounded by at least three edges, it follows, on counting up the edges around each face, that

$$3F \leq 2m$$

(the factor 2 appears since each edge bounds two faces). We obtain the required bound by replacing $F = 2 - n + m$.

(ii): This follows in a similar way, by replacing $3F \leq 2m$ with $4F \leq 2m$. $\square$

As a consequence of Corollary 1, we obtain a second, shorter proof that K_5 and $K_{3,3}$ are non-planar (Theorem 1).

Proof 2 of Theorem 1 (via Euler's Formula). $K_{3,3}$: $n = 6$, $m = 9$ (no triangles). Since $2n - 4 = 8 < m = 9$, $K_{3,3}$ is non-planar.

K_5 : $n = 5$, $m = 10$. Since $3n - 6 = 9 < m = 10$, K_5 is non-planar. $\square$

Corollary 2

Let G be a simple connected planar graph. Then G contains a vertex of degree 5 or less.

Proof. Suppose, for contradiction, that $d_i \geq 6$ for every vertex. Then

$$\sum d_i \geq 6n \implies 2m \geq 6n \implies m \geq 3n > 3n - 6,$$

contradicting $m \leq 3n - 6$ (Corollary 1(i)). □